[METRIC]

DOD-U-10866D
17 April 1979
SUPERSEDING
MIL-U-10866C
5 July 1967

MILITARY SPECIFICATION

UREA, TECHNICAL (METRIC)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two classes of technical grade urea.

1.2 Classification. Urea shall be of the following classes as specified (see 6.2):

Class 1 - For manufacturing.
Class 2 - For deicing.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-S-48 - Sacks, Shipping, Paper.

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E203 - Water Using Karl Fischer Reagent.
E258 - Total Nitrogen In Organic Materials By Modified Kjeldahl Method.
E324 - Relative Initial and Final Melting Points, and the Melting Range of Industrial Organic Chemicals.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis, Twelfth Edition, 1975.

(Application for copies should be addressed to the Association of Official Analytical Chemists, PO Box 540, Benjamin Franklin Station, Washington, DC 20044.)

(Technical society and technical associations specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Appearance.

3.1.1 Class 1. Class 1 urea shall be a white, crystalline material free from visible impurities and foreign matter when tested as specified in 4.2.4.1.

3.1.2 Class 2. Class 2 urea shall be in "shotted" or "prilled" form and shall be free flowing when tested as specified in 4.2.4.1.

3.2 Chemical and physical characteristics. Urea shall conform to the applicable chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Class 1		Class 2		Test paragraph
	Minimum	Maximum	Minimum	Maximum	
Nitrogen, percent by weight	46.0	46.7	46.0	46.7	4.2.4.2
Water, percent by weight	----	----	----	0.50	4.2.4.3
Biuret content, percent by weight	----	----	----	2.5	4.2.4.4
Ash, percent by weight	----	0.003	----	0.1	4.2.4.5
Initial melting point, °C	132.0	133.0	129.0	133.0	4.2.4.6

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of one class of urea produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 200 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled container ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing or illegible

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine the class 1 urea specimen for color, form, and the presence of impurities and foreign matter. Visually examine the class 2 urea specimen for form and flowing characteristics.

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4.2.4.2 Nitrogen. Determine percent by weight nitrogen in the specimen in accordance with ASTM E258 using the 0.5N acid procedure and a specimen weight of 0.50 to 0.60 g.

4.2.4.3 Water. Determine the percent by weight water in the specimen in accordance with ASTM E203.

4.2.4.4 Biuret. Determine the percent by weight biuret in the specimen in accordance with the procedure in chapter 2 of AOAC. Exclude the making up of the ion exchange column; prepare the standard curve using 5-milliliter (ml), 15-ml, and 25-ml quantities; and use a 2-g specimen.

4.2.4.5 Ash. Weigh to the nearest 0.01 g approximately 50 g of the specimen into an ignited, tared (to the nearest 0.1 milligram) platinum dish. Heat until a gray solid ash remains. Ignite at approximately 600°C to a constant weight. Cool to room temperature in a desiccator and weigh to the nearest 0.1 milligram. Calculate the percent by weight ash as follows:

$$\text{Percent ash} = \frac{100(A - B)}{W}$$

where: A = Weight of dish and ash in grams,
B = Tare weight of dish in grams, and
W = Weight of specimen in grams.

4.2.4.6 Initial melting point. Determine the initial melting point of the specimen in accordance with the ASTM E324 procedure for determining initial melting point.

5. PACKAGING

5.1 Packing. Urea shall be packed level A, level B, or commercial as specified (see 6.2).

5.1.1 Level A. A quantity of 36.3 (+0.36 or -0) kilograms (kg) or 45.4 (+0.45 or -0) kg as specified (see 6.2) of urea shall be packed in a multiwall paper sack conforming type optional, style B of UU-S-48 using construction No. 13 or 13X for the 36.3-kg quantity and construction No. 17 or 17X for the 45.4-kg quantity. A grade MB3 moisture barrier shall be placed next to the inner-most wall.

5.1.2 Level B. Urea shall be packed as specified in 5.1.1 except that the 36.3-kg quantity shall conform to construction No. 9 or 9X, the 45.4-kg quantity shall conform to construction No. 13 or 13X, and the moisture barrier shall have a minimum grade of MB2.

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5.1.3 Commercial. A quantity of urea as specified (see 6.2) shall be packed in accordance with MIL-STD-1188.

5.2 Marking. Level A and level B containers and palletized loads shall be marked in accordance with MIL-STD-129. Commercial containers shall be marked in accordance with MIL-STD-1188. Commercial palletized loads shall be marked in accordance with the general policies established in MIL-STD-1188 for marking shipping containers. All containers shall be marked to show the lot number and date of manufacture of the urea. Containers of class 2 urea shall be marked in bold characters to show the following information:

NOTICE

UREA, FOR USE IN DEICING AIRCRAFT RUNWAY AND RAMP AREAS ONLY. NOT AUTHORIZED FOR USE IN DEICING OF OTHER PAVEMENTS. USE ONLY AT TEMPERATURES ABOVE -9°C (15°F). SPREAD AT RATE NO GREATER THAN 1.22 KILOGRAMS PER SQUARE METER (1/4 POUND PER SQUARE FOOT).

5.3 Palletization. All containers of urea shall be palletized in uniform quantities in accordance with load type XV of MIL-STD-147 using a fiberboard cap over the load. The cap, load, and pallet shall be bonded together by reinforcement consisting of shrink-wrap, stretch-wrap, or strapping.

6. NOTES

6.1 Intended use. Class 1 urea is intended for use in the manufacture of other chemicals. Class 2 urea is intended for use in the melting of thin layers of ice at temperatures above -9°C (15°F) and for the prevention of ice accumulation on runways during freezing rain [urea is less corrosive to aircraft structural materials than rock salt (sodium chloride) and calcium chloride].

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class of urea required (see 1.2).
(c) Level of packing required (see 5.1).
(d) Unit quantity required (see 5.1.1 and 5.1.3).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

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6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

Custodians:

Army - EA
Navy - OS
Air Force - 68

Review activities:

Army - MD, SM
DLA - GS

User activities:

Army - AV
Navy - AS

Preparing activity:

Army - EA

Project No. 6810-B136

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
DOD-U-10866D (METRIC)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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